THE UNIVERSITY OF WESTERN AUSTRALIA

CSSE

SEMESTER 1, 2017 EXAMINATIONS

CITS4403
Computational Modelling

FAMILY NAME: GIVEN NAMES:

STUDENT ID: | | | | | | | | | SIGNATURE:

This Paper Contains: **16** pages **(including title page)**
Time allowed: **2:00 hours**

INSTRUCTIONS:

There are 23 questions. Marks for this paper total 120, Candidate must answer all questions.
Answers are to be written on this question paper in the spaces provided.

Calculators are not permitted.

PLEASE NOTE

Examination candidates may only bring authorised materials into the examination room. If a supervisor finds, during the examination, that you have unauthorised material, in whatever form, in the vicinity of your desk or on your person, whether in the examination room or the toilets or en route to/from the toilets, the matter will be reported to the head of school and disciplinary action will normally be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found.

Therefore, any candidate who has brought any unauthorised material whatsoever into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification.

Supervisors Only Student left at:

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There are no questions on this page.

(Questions 1 to 22 carry 5 marks each)

1. Explain how to construct a regular graph of size n and degree k when k is even (and less than n).

[5 marks]

2. Give two different methods for generating random graphs, and explain the difference in the types of graphs that they generate.

[5 marks]

3. Give pseudocode for an algorithm for determining if a graph is connected or not.

[5 marks]

4. What is the order of growth of the algorithm you described in question 3?

[5 marks]

5. Describe an efficient implementation of the Dictionary (or Map) data structure. What is the complexity of the three main operations?

[5 marks]

6. Explain what the single-source shortest path problem is and describe briefly how to solve it.

[5 marks]

7. Describe two different explanations for the Small World Phenomenon.

[5 marks]

8. Explain how the "cliquishness" of a graph can be measured by a clustering coefficient.

[5 marks]

9. What is Zipf's law?

[5 marks]

10. Do you consider Zipf's law to be an explanatory model, and why?

[5 marks]

11. Explain how to use "preferential attachment" to create random graphs with a distribution of degrees that obeys a power law.

[5 marks]

12. What is the demarcation problem? Explain how Popper proposed to address the demarcation problem.

[5 marks]

13. Define one-dimensional Cellular Automata.

[5 marks]

14. Explain why Wolfram's demonstration of complex behaviour in cellular automata is more surprising than the complexity of chaotic systems.

[5 marks]

15. In the Game of Life, there is no initial condition that yields unbounded growth in the number of live cells. True or False? Explain your answer.

[5 marks]

16. Explain how a Cellular Automaton can produce an object of dimension about 1.56.

[5 marks]

17. How are Scientific Realism and Instrumentalism related to Reductionism and Holism?

[5 marks]

18. Explain how the concept of self-organized criticality might suggest that large earthquakes are unpredictable.

[5 marks]

19. What is the fast Fourier transform, and what is its significance?

[5 marks]

20. Explain how Boids are typical agent-based models.

[5 marks]

21. Explain how you might find a good solution to the iterated prisoner's dilemma problem.

[5 marks]

22. Discuss the claim that emergence is just ignorance.

[5 marks]

23. Briefly review a chapter of a popular non-fiction book that you have read recently covering a topic related to complexity science.

[10 marks]

END OF PAPER

Blank page for extra writing if needed